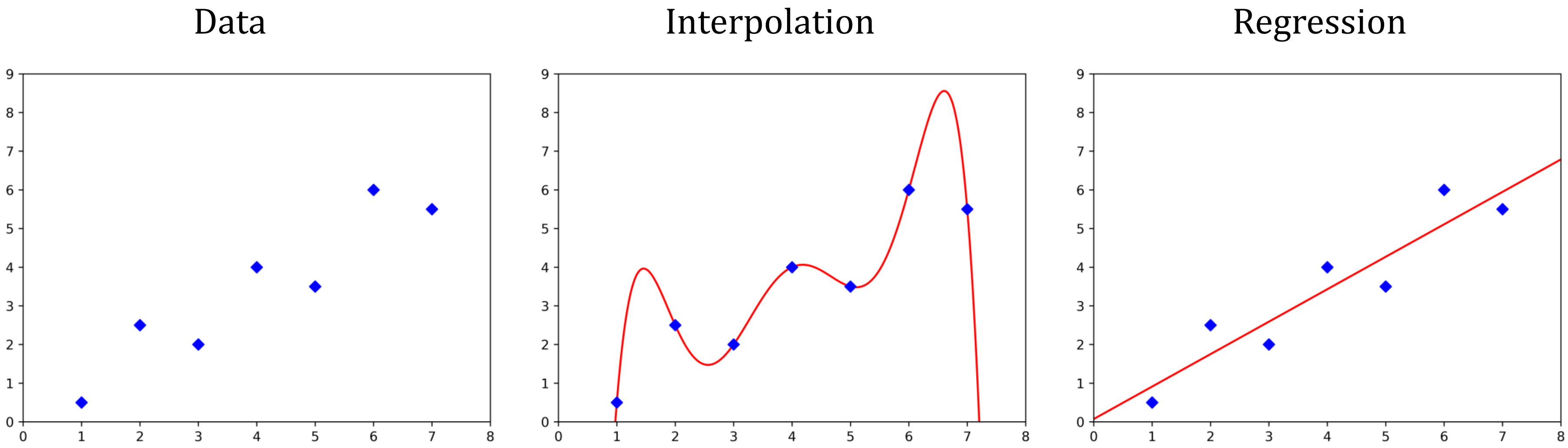
Least Squares Regression



Least Squares Regression

Linear Model

Data Set: $(x_i, y_i)|\ i = 0,1,2,\dots,n$

Linear Model: $y(x) = f(x, \mathbf{a}) = a_0 \cdot \phi_0(x) + a_1 \cdot \phi_1(x) + \cdots + a_m \cdot \phi_m(x);\quad m < n$

Error: $e_i = y_i - y(x_i) = y_i - f(x_i, \mathbf{a});\quad m < n$

Sum of Error Squares:

$$S_e = \sum_{i=0}^{n} e_i^2 = \sum_{i=0}^{n} [y_i - f(x_i, \mathbf{a})]^2;\quad m < n$$

Minimize Sum of Error Squares:

$$\frac{\partial S_e}{\partial a_0} = 0,\quad \frac{\partial S_e}{\partial a_1} = 0,\quad \dots,\quad \frac{\partial S_e}{\partial a_m} = 0;\quad m < n$$

This leads to a system of $m + 1$ equations for finding $a_0, a_1, \dots, a_m$.

$$\sum_{i=0}^{n} 2 \cdot [y_i - f(x_i)] \cdot \left(-\frac{\partial f}{\partial a_0}\right) = 0$$

$$\sum_{i=0}^{n} 2 \cdot [y_i - f(x_i)] \cdot \left(-\frac{\partial f}{\partial a_1}\right) = 0$$

$$\vdots$$

$$\sum_{i=0}^{n} 2 \cdot [y_i - f(x_i)] \cdot \left(-\frac{\partial f}{\partial a_m}\right) = 0$$

Least Squares Regression

Linear Model

Data Set: $(x_i, y_i) | \, i = 0,1,2,\ldots,n$

Linear Model: $y(x) = f(x, \mathbf{a}) = a_0 \cdot \phi_0(x) + a_1 \cdot \phi_1(x) + \cdots + a_m \cdot \phi_m(x); \quad m < n$

Error: $e_i = y_i - y(x_i) = y_i - f(x_i, \mathbf{a}); \quad m < n$

Sum of Error Squares: $S_e = \sum_{i=0}^{n} e_i^2 = \sum_{i=0}^{n} [y_i - f(x_i, \mathbf{a})]^2 \, ; \quad m < n$

Minimize Sum of Error Squares: $\frac{\partial S_e}{\partial a_0} = 0, \quad \frac{\partial S_e}{\partial a_1} = 0, \quad \ldots, \quad \frac{\partial S_e}{\partial a_m} = 0; \quad m < n$

This leads to a system of $m + 1$ equations for finding $a_0, a_1, \ldots, a_m$.

$$\sum_{i=0}^{n} 2 \cdot [y_i - f(x_i)] \cdot [-\phi_0(x_i)] = 0$$

$$\sum_{i=0}^{n} 2 \cdot [y_i - f(x_i)] \cdot [-\phi_1(x_i)] = 0$$

$$\vdots$$

$$\sum_{i=0}^{n} 2 \cdot [y_i - f(x_i)] \cdot [-\phi_m(x_i)] = 0$$

Least Squares Regression

Linear Model

Data Set: $(x_i, y_i)|\ i = 0,1,2, \dots, n$

Linear Model: $y(x) = f(x, \mathbf{a}) = a_0 \cdot \phi_0(x) + a_1 \cdot \phi_1(x) + \cdots + a_m \cdot \phi_m(x); \quad m < n$

Error: $e_i = y_i - y(x_i) = y_i - f(x_i, \mathbf{a}); \quad m < n$

Sum of Error Squares: $S_e = \sum_{i=0}^{n} e_i^2 = \sum_{i=0}^{n} [y_i - f(x_i, \mathbf{a})]^2; \quad m < n$

Minimize Sum of Error Squares: $\frac{\partial S_e}{\partial a_0} = 0, \quad \frac{\partial S_e}{\partial a_1} = 0, \quad \dots, \quad \frac{\partial S_e}{\partial a_m} = 0; \quad m < n$

This leads to a system of $m + 1$ equations for finding $a_0, a_1, \dots, a_m$.

$$\sum_{i=0}^{n} \phi_0(x_i) \cdot f(x_i) = \sum_{i=0}^{n} \phi_0(x_i) \cdot y_i$$

$$\sum_{i=0}^{n} \phi_1(x_i) \cdot f(x_i) = \sum_{i=0}^{n} \phi_1(x_i) \cdot y_i$$

$$\vdots$$

$$\sum_{i=0}^{n} \phi_m(x_i) \cdot f(x_i) = \sum_{i=0}^{n} \phi_m(x_i) \cdot y_i$$

Least Squares Regression

Linear Model

Data Set: $(x_i, y_i) | \; i = 0,1,2,\ldots,n$

Linear Model: $y(x) = f(x, \mathbf{a}) = a_0 \cdot \phi_0(x) + a_1 \cdot \phi_1(x) + \cdots + a_m \cdot \phi_m(x); \quad m < n$

Error: $e_i = y_i - y(x_i) = y_i - f(x_i, \mathbf{a}); \quad m < n$

Sum of Error Squares: $S_e = \sum_{i=0}^{n} e_i^2 = \sum_{i=0}^{n} [y_i - f(x_i, \mathbf{a})]^2 ; \quad m < n$

Minimize Sum of Error Squares: $\frac{\partial S_e}{\partial a_0} = 0, \quad \frac{\partial S_e}{\partial a_1} = 0, \quad \ldots, \quad \frac{\partial S_e}{\partial a_m} = 0; \quad m < n$

This leads to a system of $m+1$ equations for finding $a_0, a_1, \ldots, a_m$.

$$a_0 \cdot \sum_{i=0}^{n} \phi_0(x_i) \cdot \phi_0(x_i) + a_1 \cdot \sum_{i=0}^{n} \phi_0(x_i) \cdot \phi_1(x_i) + \cdots + a_m \cdot \sum_{i=0}^{n} \phi_0(x_i) \cdot \phi_m(x_i) = \sum_{i=0}^{n} \phi_0(x_i) \cdot y_i$$

$$a_0 \cdot \sum_{i=0}^{n} \phi_1(x_i) \cdot \phi_0(x_i) + a_1 \cdot \sum_{i=0}^{n} \phi_1(x_i) \cdot \phi_1(x_i) + \cdots + a_m \cdot \sum_{i=0}^{n} \phi_1(x_i) \cdot \phi_m(x_i) = \sum_{i=0}^{n} \phi_1(x_i) \cdot y_i$$

$$\vdots$$

$$a_0 \cdot \sum_{i=0}^{n} \phi_m(x_i) \cdot \phi_0(x_i) + a_1 \cdot \sum_{i=0}^{n} \phi_m(x_i) \cdot \phi_1(x_i) + \cdots + a_m \cdot \sum_{i=0}^{n} \phi_m(x_i) \cdot \phi_m(x_i) = \sum_{i=0}^{n} \phi_m(x_i) \cdot y_i$$

Least Squares Regression

Linear Model--Polynomials

$\phi_0(x) = 1, \phi_1(x) = x, \dots, \phi_m(x) = x^m$

Data Set: $(x_i, y_i) | \; i = 0,1,2, \dots, n$

Linear Model--Polynomials: $y(x) = f(x, \mathbf{a}) = P_m(x, \mathbf{a}) = a_0 + a_1 x + a_2 x^2 \dots + a_m x^m; \quad m < n$

Error: $e_i = y_i - y(x_i) = y_i - P_m(x_i, \mathbf{a}); \quad m < n$

Sum of Error Squares:

$$S_e = \sum_{i=0}^{n} e_i^2 = \sum_{i=0}^{n} [y_i - P_m(x_i, \mathbf{a})]^2 ; \quad m < n$$

Minimize Sum of Error Squares:

$$\frac{\partial S_e}{\partial a_0} = 0, \quad \frac{\partial S_e}{\partial a_1} = 0, \quad \dots, \quad \frac{\partial S_e}{\partial a_m} = 0; \quad m < n$$

This leads to a system of $m + 1$ equations for finding $a_0, a_1, \dots, a_m$.

$$\sum_{i=0}^{n} 2 \cdot [y_i - f(x_i)] \cdot \left(-\frac{\partial f}{\partial a_0}\right) = 0$$

$$\sum_{i=0}^{n} 2 \cdot [y_i - f(x_i)] \cdot \left(-\frac{\partial f}{\partial a_1}\right) = 0$$

$$\vdots$$

$$\sum_{i=0}^{n} 2 \cdot [y_i - f(x_i)] \cdot \left(-\frac{\partial f}{\partial a_m}\right) = 0$$

Least Squares Regression

Linear Model--Polynomials

$\phi_0(x) = 1, \phi_1(x) = x, \dots, \phi_m(x) = x^m$

Data Set: $(x_i, y_i) \mid i = 0,1,2, \dots, n$

Linear Model--Polynomials: $y(x) = f(x, \mathbf{a}) = P_m(x, \mathbf{a}) = a_0 + a_1 x + a_2 x^2 \dots + a_m x^m; \quad m < n$

Error: $e_i = y_i - y(x_i) = y_i - P_m(x_i, \mathbf{a}); \quad m < n$

Sum of Error Squares: $S_e = \sum_{i=0}^{n} e_i^2 = \sum_{i=0}^{n} [y_i - P_m(x_i, \mathbf{a})]^2; \quad m < n$

Minimize Sum of Error Squares: $\frac{\partial S_e}{\partial a_0} = 0, \quad \frac{\partial S_e}{\partial a_1} = 0, \quad \dots, \quad \frac{\partial S_e}{\partial a_m} = 0; \quad m < n$

This leads to a system of $m + 1$ equations for finding $a_0, a_1, \dots, a_m$.

$$\sum_{i=0}^{n} 2 \cdot [y_i - f(x_i)] \cdot (-1) = 0$$

$$\sum_{i=0}^{n} 2 \cdot [y_i - f(x_i)] \cdot (-x_i) = 0$$

$$\vdots$$

$$\sum_{i=0}^{n} 2 \cdot [y_i - f(x_i)] \cdot (-x_i^m) = 0$$

Least Squares Regression

Linear Model--Polynomials

$\phi_0(x) = 1, \phi_1(x) = x, \ldots, \phi_m(x) = x^m$

Data Set: $(x_i, y_i) |\ i = 0,1,2, \ldots, n$

Linear Model--Polynomials: $y(x) = f(x, \mathbf{a}) = P_m(x, \mathbf{a}) = a_0 + a_1 x + a_2 x^2 \ldots + a_m x^m; \quad m < n$

Error: $e_i = y_i - y(x_i) = y_i - P_m(x_i, \mathbf{a}); \quad m < n$

Sum of Error Squares: $$S_e = \sum_{i=0}^{n} e_i^2 = \sum_{i=0}^{n} [y_i - P_m(x_i, \mathbf{a})]^2 ; \quad m < n$$

Minimize Sum of Error Squares: $$\frac{\partial S_e}{\partial a_0} = 0, \quad \frac{\partial S_e}{\partial a_1} = 0, \quad \ldots, \quad \frac{\partial S_e}{\partial a_m} = 0; \quad m < n$$

This leads to a system of $m + 1$ equations for finding $a_0, a_1, \ldots, a_m$.

$$\sum_{i=0}^{n} (1) \cdot f(x_i) = \sum_{i=0}^{n} (1) \cdot y_i$$

$$\sum_{i=0}^{n} (x_i) \cdot f(x_i) = \sum_{i=0}^{n} (x_i) \cdot y_i$$

$$\vdots$$

$$\sum_{i=0}^{n} (x_i^m) \cdot f(x_i) = \sum_{i=0}^{n} (x_i^m) \cdot y_i$$

Least Squares Regression

Linear Model--Polynomials

$\phi_0(x) = 1, \phi_1(x) = x, \ldots, \phi_m(x) = x^m$

Data Set: $(x_i, y_i) | \; i = 0,1,2,\ldots,n$

Linear Model--Polynomials: $y(x) = f(x, \mathbf{a}) = P_m(x, \mathbf{a}) = a_0 + a_1 x + a_2 x^2 \ldots + a_m x^m; \quad m < n$

Error: $e_i = y_i - y(x_i) = y_i - P_m(x_i, \mathbf{a}); \quad m < n$

Sum of Error Squares:
$$S_e = \sum_{i=0}^{n} e_i^2 = \sum_{i=0}^{n} [y_i - P_m(x_i, \mathbf{a})]^2 \,; \quad m < n$$

Minimize Sum of Error Squares:
$$\frac{\partial S_e}{\partial a_0} = 0, \quad \frac{\partial S_e}{\partial a_1} = 0, \quad \ldots, \quad \frac{\partial S_e}{\partial a_m} = 0; \quad m < n$$

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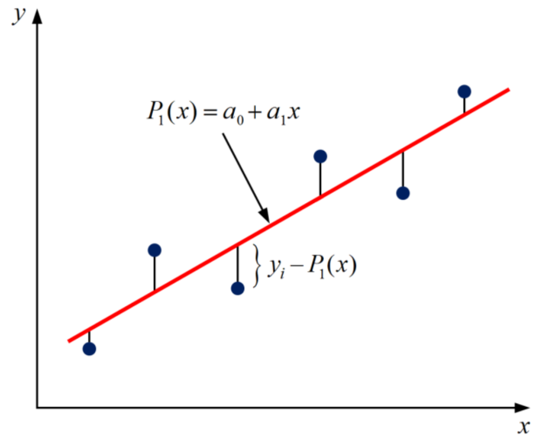
$$\sum_{i=0}^{n} (1) \cdot f(x_i) = \sum_{i=0}^{n} (1) \cdot y_i$$
$$\sum_{i=0}^{n} (x_i) \cdot f(x_i) = \sum_{i=0}^{n} (x_i) \cdot y_i$$
$$\vdots$$
$$\sum_{i=0}^{n} (x_i^m) \cdot f(x_i) = \sum_{i=0}^{n} (x_i^m) \cdot y_i$$



Least Squares Regression

Linear Model--Polynomials

Data Set: $(x_i, y_i)|\ i = 0,1,2,\ldots,n$

$\phi_0(x) = 1, \phi_1(x) = x, \ldots, \phi_m(x) = x^m$

Linear Model--Polynomials: $y(x) = f(x, \mathbf{a}) = P_m(x, \mathbf{a}) = a_0 + a_1 x + a_2 x^2 \ldots + a_m x^m; \quad m < n$

Error: $e_i = y_i - y(x_i) = y_i - P_m(x_i, \mathbf{a}); \quad m < n$

Sum of Error Squares: $S_e = \sum_{i=0}^{n} e_i^2 = \sum_{i=0}^{n} [y_i - P_m(x_i, \mathbf{a})]^2; \quad m < n$

Minimize Sum of Error Squares: $\frac{\partial S_e}{\partial a_0} = 0, \quad \frac{\partial S_e}{\partial a_1} = 0, \quad \ldots, \quad \frac{\partial S_e}{\partial a_m} = 0; \quad m < n$

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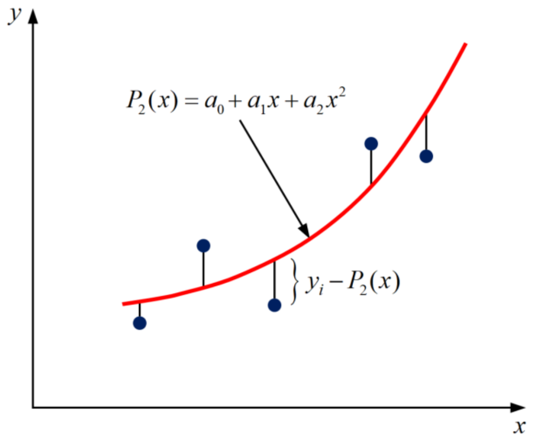
$$\sum_{i=0}^{n} (x_i) \cdot f(x_i) = \sum_{i=0}^{n} (x_i) \cdot y_i$$

$$\vdots$$

$$\sum_{i=0}^{n} (x_i^m) \cdot f(x_i) = \sum_{i=0}^{n} (x_i^m) \cdot y_i$$



Least Squares Regression

Linear Model--Polynomials

$\phi_0(x) = 1, \phi_1(x) = x, \dots, \phi_m(x) = x^m$

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Error: $e_i = y_i - y(x_i) = y_i - P_m(x_i, \mathbf{a}); \quad m < n$

Sum of Error Squares: $S_e = \sum_{i=0}^{n} e_i^2 = \sum_{i=0}^{n} [y_i - P_m(x_i, \mathbf{a})]^2 \; ; \quad m < n$

Minimize Sum of Error Squares: $\frac{\partial S_e}{\partial a_0} = 0, \quad \frac{\partial S_e}{\partial a_1} = 0, \quad \dots, \quad \frac{\partial S_e}{\partial a_m} = 0; \quad m < n$

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$$\sum_{i=0}^{n} (1) \cdot f(x_i) = \sum_{i=0}^{n} (1) \cdot y_i$$

$$\sum_{i=0}^{n} (x_i) \cdot f(x_i) = \sum_{i=0}^{n} (x_i) \cdot y_i$$

$$\vdots$$

$$\sum_{i=0}^{n} (x_i^m) \cdot f(x_i) = \sum_{i=0}^{n} (x_i^m) \cdot y_i$$

Least Squares Regression

Linear Model--Polynomials

Data Set: $(x_i, y_i)|\ i = 0,1,2, \ldots, n$

$\phi_0(x) = 1, \phi_1(x) = x, \ldots, \phi_m(x) = x^m$

Linear Model--Polynomials: $y(x) = f(x, \mathbf{a}) = P_m(x, \mathbf{a}) = a_0 + a_1 x + a_2 x^2 \ldots + a_m x^m; \quad m < n$

Error: $e_i = y_i - y(x_i) = y_i - P_m(x_i, \mathbf{a}); \quad m < n$

Sum of Error Squares: $S_e = \sum_{i=0}^{n} e_i^2 = \sum_{i=0}^{n} [y_i - P_m(x_i, \mathbf{a})]^2; \quad m < n$

Minimize Sum of Error Squares: $\frac{\partial S_e}{\partial a_0} = 0, \quad \frac{\partial S_e}{\partial a_1} = 0, \quad \ldots, \quad \frac{\partial S_e}{\partial a_m} = 0; \quad m < n$

This leads to a system of $m+1$ equations for finding $a_0, a_1, \ldots, a_m$.

$$a_0 \cdot \sum_{i=0}^{n} (1) \cdot (1) + a_1 \cdot \sum_{i=0}^{n} (1) \cdot (x_i) + \cdots + a_m \cdot \sum_{i=0}^{n} (1) \cdot (x_i^m) = \sum_{i=0}^{n} (1) \cdot y_i$$

$$a_0 \cdot \sum_{i=0}^{n} (x_i) \cdot (1) + a_1 \cdot \sum_{i=0}^{n} (x_i) \cdot (x_i) + \cdots + a_m \cdot \sum_{i=0}^{n} (x_i) \cdot (x_i^m) = \sum_{i=0}^{n} (x_i) \cdot y_i$$

$$\vdots$$

$$a_0 \cdot \sum_{i=0}^{n} (x_i^m) \cdot (1) + a_1 \cdot \sum_{i=0}^{n} (x_i^m) \cdot (x_i) + \cdots + a_m \cdot \sum_{i=0}^{n} (x_i^m) \cdot (x_i^m) = \sum_{i=0}^{n} (x_i^m) \cdot y_i$$

Least Squares Regression

Example 04-05: Find least squares regression using polynomials of order 1 for the following data.

	i	x_i	y_i
	0	1	0.5
	1	2	2.5
	2	3	2.0
	3	4	4.0
	4	5	3.5
	5	6	6.0
$n = 6$	6	7	5.5

$$P_1(x) = a_0 + a_1 x \quad (m = 1)$$

$$e_i = y_i - P_1(x_i) = y_i - a_0 - a_1 x_i$$

$$S_e = \sum_{i=0}^{6} e_i^2 = \sum_{i=0}^{6} [y_i - P_1(x_i)]^2 = \sum_{i=0}^{6} (y_i - a_0 - a_1 x_i)^2$$

$$\frac{\partial S_e}{\partial a_0} = \sum_{i=0}^{6} 2 \cdot (y_i - a_0 - a_1 x_i) \cdot (-1) = 0 \qquad a_0 \cdot \sum_{i=0}^{6} 1 + a_1 \cdot \sum_{i=0}^{6} x_i = \sum_{i=0}^{6} y_i$$

$$\frac{\partial S_e}{\partial a_1} = \sum_{i=0}^{6} 2 \cdot (y_i - a_0 - a_1 x_i) \cdot (-x_i) = 0 \qquad a_0 \cdot \sum_{i=0}^{6} x_i + a_1 \cdot \sum_{i=0}^{6} x_i^2 = \sum_{i=0}^{6} x_i y_i$$

Least Squares Regression

<u>Example 04-05</u>: Find least squares regression using polynomials of order 1 for the following data.

i	x_i	y_i
0	1	0.5
1	2	2.5
2	3	2.0
3	4	4.0
4	5	3.5
5	6	6.0
6	7	5.5

$n = 6$

$$P_1(x) = a_0 + a_1 x \quad (m = 1)$$

$$e_i = y_i - P_1(x_i) = y_i - a_0 - a_1 x_i$$

$$S_e = \sum_{i=0}^{6} e_i^2 = \sum_{i=0}^{6} [y_i - P_1(x_i)]^2 = \sum_{i=0}^{6} (y_i - a_0 - a_1 x_i)^2$$

$$\frac{\partial S_e}{\partial a_0} = \sum_{i=0}^{6} 2 \cdot (y_i - a_0 - a_1 x_i) \cdot (-1) = 0 \qquad 7a_0 + 28a_1 = 24$$

$$\frac{\partial S_e}{\partial a_1} = \sum_{i=0}^{6} 2 \cdot (y_i - a_0 - a_1 x_i) \cdot (-x_i) = 0 \qquad 28a_0 + 140a_1 = 119.5$$

$$a_0 = 0.0714286, a_1 = 0.8392857$$

$$p_1(x) = 0.0714286 + 0.8392857x$$

Least Squares Regression

Example 04-05: Find least squares regression using polynomials of order 1 for the following data.

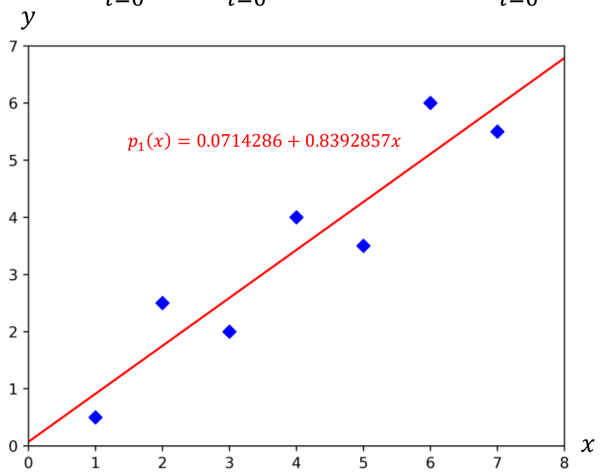
i	x_i	y_i
0	1	0.5
1	2	2.5
2	3	2.0
3	4	4.0
4	5	3.5
5	6	6.0
6	7	5.5

$n = 6$

$$P_1(x) = a_0 + a_1 x \quad (m = 1)$$

$$e_i = y_i - P_1(x_i) = y_i - a_0 - a_1 x_i$$

$$S_e = \sum_{i=0}^{6} e_i^2 = \sum_{i=0}^{6} [y_i - P_1(x_i)]^2 = \sum_{i=0}^{6} (y_i - a_0 - a_1 x_i)^2$$



$$7a_0 + 28a_1 = 24$$

$$28a_0 + 140a_1 = 119.5$$

$$a_0 = 0.0714286, a_1 = 0.8392857$$

$$p_1(x) = 0.0714286 + 0.8392857x$$

Least Squares Regression

Example 04-06: Find least squares regression using polynomials of order 2 for the following data.

	i	x_i	y_i
	0	0.00	1.0000
	1	0.25	1.2840
	2	0.50	1.6487
	3	0.75	2.1170
$n = 4$	4	1.00	2.7183

$$P_2(x) = a_0 + a_1 x + a_2 x^2 \quad (m = 2)$$

$$e_i = y_i - P_2(x_i) = y_i - a_0 - a_1 x_i - a_2 x_i^2$$

$$S_e = \sum_{i=0}^{4} e_i^2 = \sum_{i=0}^{4} [y_i - P_2(x_i)]^2 = \sum_{i=0}^{4} (y_i - a_0 - a_1 x_i - a_2 x_i^2)^2$$

$$\frac{\partial S_e}{\partial a_0} = \sum_{i=0}^{4} 2 \cdot (y_i - a_0 - a_1 x_i - a_2 x_i^2) \cdot (-1) = 0 \qquad a_0 \cdot \sum_{i=0}^{4} 1 + a_1 \cdot \sum_{i=0}^{4} x_i + a_2 \cdot \sum_{i=0}^{4} x_i^2 = \sum_{i=0}^{4} y_i$$

$$\frac{\partial S_e}{\partial a_1} = \sum_{i=0}^{4} 2 \cdot (y_i - a_0 - a_1 x_i - a_2 x_i^2) \cdot (-x_i) = 0 \qquad a_0 \cdot \sum_{i=0}^{4} x_i + a_1 \cdot \sum_{i=0}^{4} x_i^2 + a_2 \cdot \sum_{i=0}^{4} x_i^3 = \sum_{i=0}^{4} x_i y_i$$

$$\frac{\partial S_e}{\partial a_2} = \sum_{i=0}^{4} 2 \cdot (y_i - a_0 - a_1 x_i - a_2 x_i^2) \cdot (-x_i^2) = 0 \qquad a_0 \cdot \sum_{i=0}^{4} x_i^2 + a_1 \cdot \sum_{i=0}^{4} x_i^3 + a_2 \cdot \sum_{i=0}^{4} x_i^4 = \sum_{i=0}^{4} x_i^2 y_i$$

Least Squares Regression

Example 04-06: Find least squares regression using polynomials of order 2 for the following data.

i	x_i	y_i
0	0.00	1.0000
1	0.25	1.2840
2	0.50	1.6487
3	0.75	2.1170
4	1.00	2.7183

$n = 4$

$$P_2(x) = a_0 + a_1 x + a_2 x^2 \quad (m = 2)$$

$$e_i = y_i - P_2(x_i) = y_i - a_0 - a_1 x_i - a_2 x_i^2$$

$$S_e = \sum_{i=0}^{4} e_i^2 = \sum_{i=0}^{4} [y_i - P_2(x_i)]^2 = \sum_{i=0}^{4} (y_i - a_0 - a_1 x_i - a_2 x_i^2)^2$$

$$\frac{\partial S_e}{\partial a_0} = \sum_{i=0}^{4} 2 \cdot (y_i - a_0 - a_1 x_i - a_2 x_i^2) \cdot (-1) = 0 \qquad 5a_0 + 2.5a_1 + 1.875a_2 = 8.768$$

$$\frac{\partial S_e}{\partial a_1} = \sum_{i=0}^{4} 2 \cdot (y_i - a_0 - a_1 x_i - a_2 x_i^2) \cdot (-x_i) = 0 \qquad 2.5a_0 + 1.875a_1 + 1.5625a_2 = 5.4514$$

$$\frac{\partial S_e}{\partial a_2} = \sum_{i=0}^{4} 2 \cdot (y_i - a_0 - a_1 x_i - a_2 x_i^2) \cdot (-x_i^2) = 0 \qquad 1.875a_0 + 1.5625a_1 + 1.382813a_2 = 4.401538$$

$$a_0 = 1.0051373, a_1 = 0.8641814, a_2 = 0.8436586$$

$$p_2(x) = 1.0051373 + 0.8641814x + 0.8436586x^2$$

Least Squares Regression

Example 04-06: Find least squares regression using polynomials of order 2 for the following data.

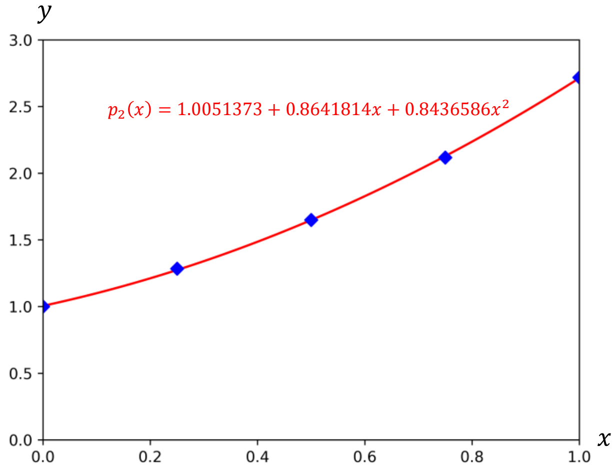
i	x_i	y_i
0	0.00	1.0000
1	0.25	1.2840
2	0.50	1.6487
3	0.75	2.1170
4	1.00	2.7183

$n = 4$

$$P_2(x) = a_0 + a_1 x + a_2 x^2 \quad (m = 2)$$

$$e_i = y_i - P_2(x_i) = y_i - a_0 - a_1 x_i - a_2 x_i^2$$

$$S_e = \sum_{i=0}^{4} e_i^2 = \sum_{i=0}^{4} [y_i - P_2(x_i)]^2 = \sum_{i=0}^{4} (y_i - a_0 - a_1 x_i - a_2 x_i^2)^2$$



$$5a_0 + 2.5a_1 + 1.875a_2 = 8.768$$

$$2.5a_0 + 1.875a_1 + 1.5625a_2 = 5.4514$$

$$1.875a_0 + 1.5625a_1 + 1.382813a_2 = 4.401538$$

$$a_0 = 1.0051373, a_1 = 0.8641814, a_2 = 0.8436586$$

$$p_2(x) = 1.0051373 + 0.8641814x + 0.8436586x^2$$

Least Squares Regression

Example 04-07: Find least squares regression using polynomials of order 1 passing origin for the following data.

i	x_i	y_i
0	1	0.5
1	2	2.5
2	3	2.0
3	4	4.0
4	5	3.5
5	6	6.0
6	7	5.5

$n = 6$

$$P_1(x) = a_1 x \quad (m = 1, a_0 = 0)$$

$$e_i = y_i - P_1(x_i) = y_i - a_1 x_i$$

$$S_e = \sum_{i=0}^{6} e_i^2 = \sum_{i=0}^{6} [y_i - P_1(x_i)]^2 = \sum_{i=0}^{6} (y_i - a_1 x_i)^2$$

$$\frac{\partial S_e}{\partial a_1} = \sum_{i=0}^{6} 2 \cdot (y_i - a_1 x_i) \cdot (-x_i) = 0$$

$$a_1 \cdot \sum_{i=0}^{6} x_i^2 = \sum_{i=0}^{6} x_i y_i$$

$$a_1 = \frac{119.5}{140} = 0.8535714$$

$$p_1(x) = 0.8535714x$$